Besag2 model for weighted spatial effects

Parametrization

The besag2 model is an extention to the besag model. Let the random vector $\mathbf{z} = (x_1, \ldots, x_n)$ be the besag model, then the besag2 is the following extentions

$$\mathbf{x} = (a\mathbf{z}, \mathbf{z}/a)$$

where $a > 0$ is an additional hyperparameter and $\dim(\mathbf{x}) = 2n$, and $\mathbf{z}$ is the *same* (up to tiny additive noise) random vector.

Hyperparameters

This model has two hyperparameters $\theta = (\theta_1, \theta_2)$.

The precision parameter τ is represented as

$$\theta_1 = \log \tau$$

and the prior is defined on θ_1.

The weight-parameter a is represented as

$$\theta_2 = \log a$$

and the prior is defined on θ_2.

Specification

The besag2 model is specified inside the `f()` function as

```
f(<whatever>,model="besag2", graph=<graph>
              precision=<precision>, hyper = <hyper>,
              adjust.for.con.comp = TRUE,
              constr=TRUE,
              scale.model = FALSE)
```

The precision is the precision defining how equal the two copies of $\mathbf{z}$ is. The neighbourhood structure of $\mathbf{x}$ is passed to the program through the `graph` argument.

Note that the besag2 model has dimension $2n$, where n is the size of the graph.

If the option `adjust.for.con.comp=TRUE` then the model is adjusted if the graph has more than one connected compoment. This adjustment can be disabled setting this option to `FALSE`. If `adjust.for.con.comp=TRUE` then `constr=TRUE` is interpreted as a sum-to-zero constraint on *each* connected component in the graph and the `rankdef` parameter is set depending on the number of connected components.

The logical option `scale.model` determine if the model z should be scaled to have an average variance (the diagonal of the generalized inverse) equal to 1. This makes prior spesification much easier. For historical reasons, the default is `FALSE` so that the model is not scaled, but its is **HIGHLY RECOMMENDED** to set this option to `TRUE`.

Hyperparameter spesification and default values

doc `The shared Besag model`

hyper

theta1

hyperid 9001
name log precision
short.name prec
prior loggamma
param 1 5e-05
initial 4
fixed FALSE
to.theta function(x) log(x)
from.theta function(x) exp(x)

theta2

hyperid 9002
name scaling parameter
short.name a
prior loggamma
param 10 10
initial 0
fixed FALSE
to.theta function(x) log(x)
from.theta function(x) exp(x)

constr TRUE

nrow.ncol FALSE

augmented FALSE

aug.factor 1

aug.constr 1 2

n.div.by 2

n.required TRUE

set.default.values TRUE

pdf besag2

Example

This is a simulated example.

```
data(Oral)
g = system.file("demodata/germany.graph", package="INLA")

## use data Oral to estimate a spatial field in order to simulate a
## 'realistic' dataset.
formula = Y ~ f(region, model="bym", graph=g)
result = inla(formula, data = Oral, family = "poisson", E = E)

x = result$summary.random$region$mean
n = length(x)/2
```

```
## simulate two new datasets. 'a' is the scaling between the
## log.rel.risk:
a = 2
xx = x[1:n]+1
x = c(0 + a*xx, 1 + xx/a)
E = c(Oral$E, Oral$E)
N = 2*n
y = rpois(N, lambda = E*exp(x))

## model='besag2' defines a model with length N = 2*graph->n, the
## first half is weighted with 'a' the other half is weighted with
## 1/a. here there is no unstructed terms.
idx = 1:N
mu = as.factor(rep(1:2, each=n))
formula = y ~ -1 + mu + f(idx, model="besag2", graph=g, scale.model=TRUE)
r = inla(formula, family = "poisson", data = data.frame(E,y,idx,mu), E=E, verbose=TRUE)
```

Details on the implementation

This gives some details of the implementation, which depends on the following variables

nc1 Number of connected components in the graph with size 1. These nodes, *singletons*, have no neighbours.

nc2 Number of connected components in the graph with size ≥ 2.

scale.model The value of the logical flag, if the model should be scaled or not. (Default FALSE)

adjust.for.con.comp The value of the logical flag if the `constr=TRUE` option should be reinterpreted.

The case `(scale.model==FALSE && adjust.for.con.comp == FALSE)`

The option `constr=TRUE` is interpreted as a sum-to-zero constraint over the whole graph. Singletons are given a uniform distribution on $(-\infty, \infty)$ before the constraint.

The case `(scale.model==TRUE && adjust.for.con.comp == FALSE)`

The option `constr=TRUE` is interpreted as a sum-to-zero constraint over the whole graph. Let $Q = \tau R$ be the standard precision matrix from the `besag`-model with precision parameter τ. Then R, except the singletons, are scaled so that the geometric mean of the marginal variances is 1, and R is modified so that singletons have a standard Normal distribution.

The case `(scale.model==FALSE && adjust.for.con.comp == TRUE)`

The option `constr=TRUE` is interpreted as one sum-to-zero constraint over each of the `nc2` connected components of size ≥ 2. Singletons are given a uniform distribution on $(-\infty, \infty)$.

The case `(scale.model==TRUE && adjust.for.con.comp == TRUE)`

The option `constr=TRUE` is interpreted as `nc2` sum-to-zero constraints for each of the connected components of size ≥ 2. Let $Q = \tau R$ be the standard precision matrix from the `besag`-model with

precision parameter τ. Then R, are scaled so that the geometric mean of the marginal variances in each connected component of size ≥ 2 is 1, and modified so that singletons have a standard Normal distribution.

Notes